

Fix: Harden input handling (protobuf.js)

Fix: Harden input handling (protobuf.js) - dcodeIO, protobuf.js.

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Content

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fix: Harden input handling

- Repository: protobufjs/protobuf.js
- Opened by: dcodeIO
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- State: closed

Body

Tightens validation and code generation handling for input edge cases, and refreshes generated fixtures and TypeScript definitions.

Comments

Xvush, 2026-04-27

reviewed the utf8 hardening. the 2-byte, 3-byte, and 4-byte overlong cases are all rejected correctly against the GHSA scenarios -- path traversal (0xC0 0xAF), NUL injection (0xC0 0x80, 0xE0 0x80 0x80), overlong slash via E0/F0 lead (0xE0 0x81 0xAF, 0xF0 0x80 0x80 0xAF) all return U+FFFD now.

two observations on edge cases that the current diff does not cover:

1. lone surrogates in the 3-byte branch. input `0xED 0xA0 0x80` decodes to U+D800 (a high surrogate that should not appear standalone in well-formed UTF-8). RFC 3629 §3 forbids encoding U+D800-U+DFFF. the current `c3 >= 0x800` check rejects below-U+0800 overlongs but does not reject the surrogate range. concrete:

```
const utf8 = require('@protobufjs/utf8');
const t = Buffer.from([0xED, 0xA0, 0x80]);
console.log(utf8.read(t, 0, t.length).charCodeAt(0).toString(16)); // "d800"
```

minimal patch alongside the existing `c3 >= 0x800` :

```
- str += c3 >= 0x800 ? String.fromCharCode(c3) : replacementChar;
+ str += (c3 >= 0x800 && (c3 < 0xD800 || c3 > 0xDFFF))
+   ? String.fromCharCode(c3)
+   : replacementChar;
```

2. above-U+10FFFF in the 4-byte branch. input `0xF4 0x90 0x80 0x80` decodes to U+110000, which is outside the Unicode codespace (max U+10FFFF). the current `t2 < 0x10000` check catches 4-byte-encoded values that should have been encoded in fewer bytes, but does not catch values that exceed the Unicode max. the bit math at line 64 (`String.fromCharCode(0xDC00 + (t2 & 0x3FFF))`) still emits surrogates without bounds checking.

minimal patch:

```
- if (t2 < 0x10000)
+ if (t2 < 0x10000 || t2 > 0x10FFFF)
+   str += replacementChar;
+ else {
```

i can send the regression test cases i drafted in the private fork (commit `366d608` on `fix/overlong-utf8-decode` , 53/53 pass via `tape tests/util_utf8.js`) as a follow-up commit if useful for coverage. the suite covers the four GHSA scenarios (path traversal, XSS, SQLi, NUL injection), per-class lead-byte assertions (`0xC0/0xC1`, 3-byte overlong, surrogates `0xED`, `> U+10FFFF`), 7-case cross-conformance with `Buffer.toString("utf8")` , and a separate fallback-path test that forces the manual decoder by passing plain `number[]` . the existing fixture at `tests/data/util_utf8/utf8.txt` covers roundtrip on valid input but does not exercise these invalid-byte paths.

nothing in this PR contradicts the report's findings on the 2/3/4-byte overlong rejection -- the approach is cleaner than my draft. just flagging the two edge cases above for completeness before merge.

if you'd prefer, i can open a follow-up PR against `patch/input-hardening` with these two edge-case fixes + the regression test suite -- your call on whether it's cleaner to land here or in a separate PR.

Tofandel, 2026-04-28

This seems to be broken

```
protos:build:protos: > pbjs --force-message --null-semantics -t static-module -w wrapper.js --force-number --dependency protobufjs/minimal.js --es6 -o ./protos.js ../../..../protos/*
protos:build:protos: node_modules/protobufjs-cli/pbjs.js:256 protos:build:protos: throw err;
protos:build:protos: ^
```

protos:build:protos: protos:build:protos: TypeError: Cannot read properties of undefined (reading 'reservedRe')

Edit: version mismatch between protobufjs-cli and protobufjs. Some version constraints when making incompatible changes would be welcome

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